

Nifty 100 Financial Analytics Platform

Comprehensive Financial Analyst & Developer Operations Guide

Version: 1.0.0 | **Release Date:** August 2026

1. Platform Architecture & Data Pipeline

The Nifty 100 Financial Analytics Platform standardizes fundamentals across 92 non-financial and financial constituent equities. Data is processed through SQLite database engines, enriched with NLP parsing for CAGR/ROE commentary, evaluated via 12-rule auto pros/cons engines, and served via FastAPI REST services.

Operational Details & Commands:

- Run pytest suite: `pytest tests/ -v --html=reports/pytest_report.html`
- Start API server: `uvicorn src.api.main:app --port 8000`
- Generate Tearsheets: `python src/reports/batch_generate.py`

2. Interactive Streamlit Screener Guide

How to navigate the 6 preset screener views: Quality Compounders, Value Picks, Growth Accelerators, Dividend Champions, Debt-Free Blue Chips, and Turnaround Watch. Filter metrics include ROE, Debt/Equity, FCF, and multi-year CAGR thresholds.

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3. Company Profile & Financial Statements

Step-by-step navigation of 10-year historical Profit & Loss statements, Balance Sheets, Cash Flow statements, and computed Financial Ratios. Metrics feature DuPont ROE decompositions and trend signals.

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4. Executive PDF Tearsheets

How PDF tearsheets are auto-generated on-demand using ReportLab. Key visual elements include KPI Tiles, 10-Year Revenue/PAT bar charts, ROE/ROCE line charts, Balance Sheet stacked bars, Cash Flow waterfalls, and auto-generated Pros/Cons.

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5. REST API Developer Documentation

Detailed walkthrough of all 16 FastAPI endpoints mounted at /api/v1 prefix, including /health, /companies, /screener, /sectors, /peers, /valuation, and /tearsheet PDF binary streaming.

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6. KMeans Clustering & Archetype Profiles

Understanding the 5 machine learning archetypes: High-Quality Compounders, Defensive Dividend Payers, Value Cyclicals, Emerging Growth, and Distressed/Turnaround. Distance metrics measure deviation from cluster centroids.

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7. Cash Flow Intelligence & CFO Quality Metrics

Evaluating CFO Quality Score (CFO/PAT) and CapEx Intensity (|CFI|/Sales). Distress alerts flag negative CFO alongside positive financing cash flows.

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8. Capital Allocation & YoY Pattern Tracking

Categorization across 8 capital allocation strategies including Shareholder Return Machines, Heavy Reinvestors, and Deleveraging Compounders.

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9. Automated Unit Testing & Data Quality Checks

Executing the pytest suite with 100+ unit and integration tests. Rules verify non-negative revenues, valid ratio boundaries, and foreign key integrity.

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10. Deployment, Performance Tuning & Troubleshooting

Uvicorn server execution, concurrent thread benchmarks, SQLite indexing optimizations, and common environment troubleshooting steps.

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